European Food Research and Technology
https://doi.org/10.1007/s00217-023-04373-z

ORIGINAL PAPER

Characterization and discrimination of camellia oil varieties according to fatty acid, squalene, tocopherol, and volatile substance contents by chromatography and chemometrics

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Received: 29 June 2023 / Revised: 4 September 2023 / Accepted: 9 September 2023
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Abstract
Adulteration of camellia oils from the same source should be avoided. Hence, using chromatographic techniques, this study aimed to analyze and select components (fatty acids, squalene, tocopherols, and volatile substances) as variables for the precise characterization and discrimination of seven camellia oil varieties (*Camellia gigantocarpa*, *Camellia vietnamensis*, *Camellia osmantha*, *Camellia nanyongensis*, *Camellia oleifera*, *Camellia semiserrata*, and *Camellia amplexicaulis*). The principal component and hierarchical cluster analyses were performed to identify camellia seed oils, and their identification ability for different components was compared. All the components, except fatty acids, could clearly distinguish camellia oil varieties. However, when the fatty acids were combined with either squalene or tocopherol, the identification ability significantly improved. Identifying different varieties of camellia oils according to the abovementioned components provides a solid foundation for the fine processing of camellia oils, and the corresponding products should be developed according to these characteristics to enhance their value.

Keywords Camellia oil · Fatty acids · Tocopherol · Squalene · Volatile substances

Introduction

Camellia oils, also known as tea-seed oils or oil-tea camellia seed oils, are mainly composed of triacylglycerols and some lipid phytochemicals, such as tocopherols, squalene, and phytosterols [1]. They are a vital ingredient in our diet and could provide energy and essential nutrients for the human body. Historically, these seeds have been cultivated and consumed in Asia, especially in China [1, 2]. Being similar to olive oil in terms of fatty acid and lipid phytochemical contents, the camellia oil is also called "Oriental olive oil" [2, 3].

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Camellia oil and olive oil are regarded as luxury edible oils in the market, but they have also become the target of shoddy [4]. Many researchers are committed to studying the differences in fatty acids, sterol, and tocopherol contents and other aspects between camellia seed oil and other edible vegetable oils. For example, Zhang et al. [5, 6] distinguished five edible oils (peanut, rapeseed, sesame, sunflower seed, and soybean oils) through principal component analysis (PCA) and hierarchical cluster analysis (HCA) as multivariate statistical methods. Troya et al. [7] classified six vegetable oils (grape seed, olive, peanut, hazelnut, sunflower seed and corn oils) according to their plant sources by using *n*-alkane spectra and Linear Discriminant Analysis models. Furthermore, a method for distinguishing virgin olive oil from other edible oils was presented by Bagur-Gonzalez et al. [8], which combining chemometrics and liquid chromatography fingerprints of sterols. These methods can identify different kinds of edible vegetable oils, but they are not necessarily effective for several varieties of edible vegetable oils from the same species. The yield, quality, and price greatly differ between camellia seed oil varieties [9]. Camellia oil processors cannot classify camellia seed varieties in detail. Although the products are not inspected in terms of

Published online: 05 October 2023

adulteration, they still show reduced quality of camellia oils and affect the brand building of camellia oils. The main reason is the low degree of refined utilization of camellia seeds. Some camellia oils with low unsaturated fatty acid content or low lipid concomitant content are not ideal as edible oil, but they can be used as cosmetic oil. Unlike soybeans and rapeseed, woody oil plants such as camellia seeds and olives require a long period from planting to harvesting. Farmers' income can only be increased by optimizing the value of existing camellia seeds. However, the quality differences of camellia oil varieties remain insufficiently known.

This study aimed to compare the ability of various components of camellia oil in distinguishing camellia oil varieties and to preliminarily classify camellia oils for fine processing and utilization. We selected the following seven representative camellia oil seed varieties in Guangxi, China: *Camellia gigantocarpa*, *Camellia vietnamensis*, *Camellia osmantha*, *Camellia nanyongensis*, *Camellia oleifera*, *Camellia semiserrata*, and *Camellia amplexicaulis*. The data of fatty acids, squalene, tocopherols, and volatile substances were obtained by gas chromatography (GC), high-performance liquid chromatography (HPLC), and gas chromatography with mass spectrometry (GC–MS) and were identified and classified using PCA and HCA.

Materials and methods

Materials and chemical samples

All the seven camellia oil varieties were collected from the camellia seed base of Guangxi Forestry Research Institute. A small expeller equipped with an internal cooling system was used to press the seed samples and ensure that the crude oil temperature remained below 80 °C. The pitch was adjusted to maintain the thickness of the camellia bran between 1.5 and 2.0 mm, and after the pressing was completed, filtering was performed.

Standards and reagents

We purchased reference standard GLC NESTLE 37 MIX from Solarbic (Beijing, China), as well as squalene (≥ 98%) and α-, β-, γ-, and δ-tocopherol standards (HPLC grade, ≥ 95%) from Macklin (Shanghai, China).

Fatty acid determination

Sample preparation

Following the method outlined by Shen et al. [10], we accurately weighed 2.0 g of oil sample in a flat-bottom flask, added 10 mL of 2 M methanolic potassium hydroxide (KOH) solution and 1 mL of methanol solution, and saponified the mixture at 85 °C. The saponification solution was washed several times with saturated salt water and 50 mL of dichloromethane until the lower liquid was clear and transparent. Next, we added and centrifuged anhydrous sodium sulfate at 4400 r/min for 10 min. The supernatant containing fatty acid methyl esters (FAMEs) was separated and then filtered using a polytetrafluoroethylene (PTFE) syringe filter (0.22 μm) for GC analysis. All analyses were repeated thrice.

GC analysis

To determine the fatty acids, a Shimadzu 2010 GC equipped with an SGE capillary column (30 m × 0.32 mm × 0.25 μm) was used following the method described by Shi Ting et al. [11]. The injector was programmed to run using a splitless injection method, with an injector temperature of 250 ℃ and an injection volume of 1.0 μL. The column flow rate was 1.0 mL/min (ultra-high–purity [UHP] nitrogen). The initial temperature of the oven was set to 60 ℃ and maintained for 1 min, then increased to 170 ℃ at 5 ℃/min, held for 2 min, increased further to 220 ℃ at 10 ℃/min, and finally held for 5 min. The FAME peaks in the oil samples were identified qualitatively based on their retention times compared to those of the reference standards. Individual fatty acids were quantified using the AOCS Official Method Ce 1 h-05, and their amount was calculated using the theoretical flame ionization detector correction factor (TCF).

Tocopherol determination

Sample preparation

We accurately weighed and placed 3 g samples in a centrifuge tube and filtered using PTFE syringe filters (0.22 μm). Thereafter, we directly injected the sample solutions into the HPLC instrument [5, 6, 12]. All the analyses were conducted thrice.

HPLC analysis

The Agilent 1260 HPLC system with a fluorescence detector (excitation, 295 nm; emission, 325 nm) was used to analyze tocopherols. An NH_2 column (250 × 4.0 mm, 5 μm; Agilent Technologies, USA) was utilized for separating the components. The detection parameters were as follows: mobile phase, hexane/isopropanol at a ratio of 98:2 (v/v). The following parameters were used: flow rate of 1.0 mL/min, injection volume of 10 μL, column temperature of 35 ℃, and a total run time of 20 min. Additionally, the external standard method was used to quantify the content of tocopherol by analyzing the mixed standard solution of tocopherol at 10–150 mg/L and obtaining a calibration curve.

Squalene determination

Sample preparation

As Shen et al. [10] previously described, 300 μL of the internal standard solution of squalene (1 mg/mL) was added into a 250 mL round-bottom flask and dried using a nitrogen blow dryer. Next, 1.0 g of the sample (accurate to 0.0001 g) was added to 50 mL of KOH-ethanol solution (1 mol/L) and then saponified at 80℃ for 50 min. Subsequently, we stopped the heating and added 50 mL of water from the top of the reflux tube. The flask was then removed, well-shaken, and cooled at room temperature. We transferred the saponified liquid to a 250 mL separating funnel and washed the flask thrice with 50 mL of *n*-hexane. The washing liquid was poured into the separating funnel, which was then shaken for 2 min. After standing and stratification, the lower saponified liquid was transferred to another 250 mL separating funnel. The abovementioned operation was repeated thrice, and the extract was combined in the same separating funnel. Furthermore, the *n*-hexane extract was washed three to four times using 25 mL of 10% ethanol solution, and the lower ethanol aqueous solution was discarded each time and tested with pH test paper until the lower effluent became neutral. The neutral *n*-hexane extract was filtered into a spherical evaporation bottle through a filter paper coated with approximately 5 g of anhydrous sodium sulfate. Afterward, the separation funnel and anhydrous sodium sulfate were rinsed thrice with approximately 20 mL of *n*-hexane. The *n*-hexane was recovered in a 40 ℃ water bath. When roughly 2 mL of *n*-hexane was left in the bottle, the evaporation bottle was removed, and the nitrogen was blown to dry. We then dissolved the dry extract in 10 mL of *n*-hexane and filtered this solution through a 0.22 mm membrane filter unit before being analyzed by GC–MS.

GC analysis

To determine the amount of squalene, Shimadzu 2010 GC with an HP-5 capillary column (30 m × 0.32 mm × 0.25 μm) was used and the method described by Shen et al. [10] was followed. The injector operated in a split injection mode with a ratio of 1:10 at 250 ℃. The carrier gas flow rate was set to 1.0 mL/min (UHP nitrogen). The oven temperature was initially set at 160 ℃, and then increased to 220 ℃ at a rate of 15 ℃/min. It was held at this temperature for 2 min, then the temperature was further increased to 280 ℃ at a rate of 5 ℃/min and held for 20 min. Finally, the temperature was increased to 300 ℃ at 5 ℃/min and held for 2 min. The detector temperature was set at 300 ℃.

Volatile substance determination

Sample preparation

We accurately weighed 2 g of the oil samples into a 15 mL headspace injection bottle. The samples were extracted at 60 ℃ for 45 min using DVB/PDMS/CarbonWR (80 μm) and analyzed at 250 ℃ for 3 min [13].

GC–MS analysis

The Shimadzu 8050 GC–MS with an InerCap 5MS/Sil capillary column (30 m × 0.32 mm × 0.5 μm) was used to identify volatile substances. The injector was set to function in the splitless mode (that is, the whole sample was injected), and the temperature was maintained at 250 ℃. The carrier gas flow rate was set to 1.0 mL/min (UHP helium). The oven temperature was initially set at 50 ℃ for 2 min, then increased to 90 ℃ at a rate of 1 ℃/min and further increased to 250 ℃ at a rate of 10 ℃/min. It was held at this temperature for 10 min. Additionally, the MS was operated in an electron impact mode with an electron energy of 70 eV. The ion source, transfer line, and quadrupole temperatures were set at 230 ℃, 280 ℃, and 150 ℃, respectively [14].

Multivariate statistical analysis

PCA reduces the dimensionality of the original data, calculates its covariance matrix and the eigenvalues and eigenvectors of the covariance matrix, and selects components with a cumulative variance contribution rate of ≥ 85% as principal components to evaluate the similarities and differences between samples. HCA mainly uses K-means, which employs distance as the evaluation indicator of similarity, because closer the distance between two objects, greater their similarity. The calculation processes of PCA and HCA were performed using OriginPro (version 2021, Massachusetts, USA).

Results and discussion

Characterization and discrimination of camellia oils based on fatty acids data

Table 1 lists 21 fatty acids and summarizes the fatty acids contents (g/100 g) of 7 camellia oil samples in comparison with the retention time of the reference standards. Accordingly, palmitic acid (C16:0), palmitoleic acid (C16:1), stearic acid (C18:0), oleic acid (C18:1n9c), linoleic acid (C18:2n6c), arachidic acid (C20:0), and nervonic acid (C24:1) were detected in all samples. They are the main fatty acids in oils [15, 16]. However, oils extracted

Table 1 Fatty acids content in camellia oils

Camellia oils		*C. gigantocarpa*	*C. vietnamensis*	*C. osmantha*	*C. nanyongensis*	*C. oleifera*	*C. semiserrata*	*C. amplexicaulis*
Fatty acids content (g/100 g)	C6:0	ND	ND	ND	0.02 ± 0.01	ND	ND	ND
	C13:0	ND	ND	ND	0.02 ± 0.01	ND	ND	ND
	C14:0	0.09 ± 0.02	0.04 ± 0.01	0.04 ± 0.01	0.05 ± 0.01	0.04 ± 0.01	0.07 ± 0.02	ND
	C15:0	ND	ND	ND	0.01 ± 0.00	ND	0.01 ± 0.00	ND
	C16:0	14.30 ± 0.01	12.00 ± 0.01	11.10 ± 0.01	9.89 ± 0.01	8.47 ± 0.01	9.12 ± 0.01	45.90 ± 0.01
	C16:1	0.06 ± 0.01	0.09 ± 0.01	0.09 ± 0.01	0.06 ± 0.01	0.11 ± 0.01	0.10 ± 0.01	0.15 ± 0.01
	C17:0	0.05 ± 0.01	0.06 ± 0.01	0.06 ± 0.01	0.09 ± 0.01	0.06 ± 0.01	0.09 ± 0.01	ND
	C17:1	ND	ND	ND	ND	ND	ND	0.04 ± 0.01
	C18:0	3.15 ± 0.01	2.02 ± 0.01	2.86 ± 0.01	3.04 ± 0.01	2.01 ± 0.05	2.10 ± 0.03	2.42 ± 0.04
	C18:1n9c	66.46 ± 0.01	72.80 ± 0.01	75.90 ± 0.01	75.30 ± 0.01	78.50 ± 0.10	74.60 ± 0.12	48.20 ± 0.15
	C18:2n6c	13.80 ± 0.01	10.80 ± 0.01	8.05 ± 0.01	8.60 ± 0.01	8.77 ± 0.05	11.50 ± 0.20	1.83 ± 0.03
	C20:0	0.06 ± 0.01	0.04 ± 0.01	0.05 ± 0.01	0.05 ± 0.01	0.04 ± 0.01	0.04 ± 0.01	0.06 ± 0.01
	C20:1	0.60 ± 0.01	0.26 ± 0.01	0.40 ± 0.01	ND	0.31 ± 0.01	0.44 ± 0.02	0.11 ± 0.02
	C18:3n3	0.45 ± 0.01	0.44 ± 0.01	0.39 ± 0.01	ND	0.48 ± 0.01	0.55 ± 0.01	0.25 ± 0.03
	C22:0	0.02 ± 0.01	0.01 ± 0.01	0.02 ± 0.01	0.02 ± 0.01	0.01 ± 0.00	0.02 ± 0.00	ND
	C20:3n3	0.03 ± 0.01	0.03 ± 0.01	0.02 ± 0.01	0.04 ± 0.01	0.03 ± 0.01	0.03 ± 0.00	0.01 ± 0.00
	C23:0	0.01 ± 0.01	ND	ND	0.09 ± 0.01	ND	ND	0.01 ± 0.00
	C22:2	ND	ND	ND	ND	ND	ND	0.01 ± 0.00
	C24:0	ND	ND	ND	0.02 ± 0.01	ND	ND	ND
	C20:5n3	0.05 ± 0.01	0.03 ± 0.01	0.03 ± 0.01	0.05 ± 0.01	0.03 ± 0.01	0.05 ± 0.00	0.03 ± 0.00
	C24:1	0.05 ± 0.01	0.06 ± 0.01	0.04 ± 0.01	0.09 ± 0.01	0.06 ± 0.01	0.08 ± 0.01	0.02 ± 0.00

The notation "ND" was used to indicate that the substance was not detected at the limits of quantitation. To facilitate chemometric analysis, the content% of samples with the result "ND" was considered 0

[a]Data are expressed as mean ± standard deviation (SD)

from different camellia oil seeds have different fatty acid compositions. The *C. amplexicaulis* oil has the highest levels of palmitic acid (45.90 g/100 g) and palmitoleic acid (0.15 g/100 g), but its oleic acid (48.20 g/100 g) and linoleic acid (1.83 g/100 g) contents are quite lower than those in other oils. Conversely, *C. oleifera* oil has the lowest level of palmitic acid (8.47 g/100 g) but has the highest level of oleic acid among all varieties (78.50 g/100 g), consistent with the study result of Cao Yongqing et al. [17]. Moreover, *C. semiserrata* oil is rich in oleic acid (74.60 g/100 g) and linoleic acid (11.50 g/100 g), which is slightly different from that reported in the study by Zhong et al. [18] probably because the samples were from different regions. Meanwhile, *C. semiserrata* and *C. oleifera* oils have similar fatty acid compositions. The linoleic acid (13.80 g/100 g) and palmitic acid (14.30 g/100 g) contents of *C. gigantocarpa* oil are also high, but the oleic acid content (66.46 g/100 g) is quite lower than that in other varieties apart from *C. amplexicaulis*. In addition, the oleic acid (75.90 g/100 g) and linoleic acid (8.05 g/100 g) contents of *C. osmantha* oil are quite similar to those of *C. nanyongensis* oil (75.30 g/100 g and 8.60 g/100 g, respectively).

This study sought to characterize and distinguish the several varieties of camellia oil. Therefore, common chemometrics methods, including PCA and HCA, were utilized, and fatty acids data were used as variables to establish differences between such varieties. Considering the source of the samples, the judgment of discrimination of camellia oil varieties by one-way analysis of variance could not be made before the classification analysis.

Based on the data matrix presented in Table 1 (21 variables × 7 samples), the ability of fatty acids to distinguish camellia oil varieties was determined using PCA and the variables that were minimized to explain most of the variation in the original data. The Kaiser rule was applied to determine the number of principal components (PCs) extracted from the variables. The gravel test, which only included PCs with eigenvalues of > 1, was performed. This criterion ensures that only PCs with eigenvalues of > 1 are retained. The PCA results showed that two PCs were extracted from the fatty acid data, explaining 84.24% of the variance. The PC1 and PC2 score plots revealed that different camellia oils were mainly grouped according to their variety sources. As shown in the PCA results (Fig. 1a),

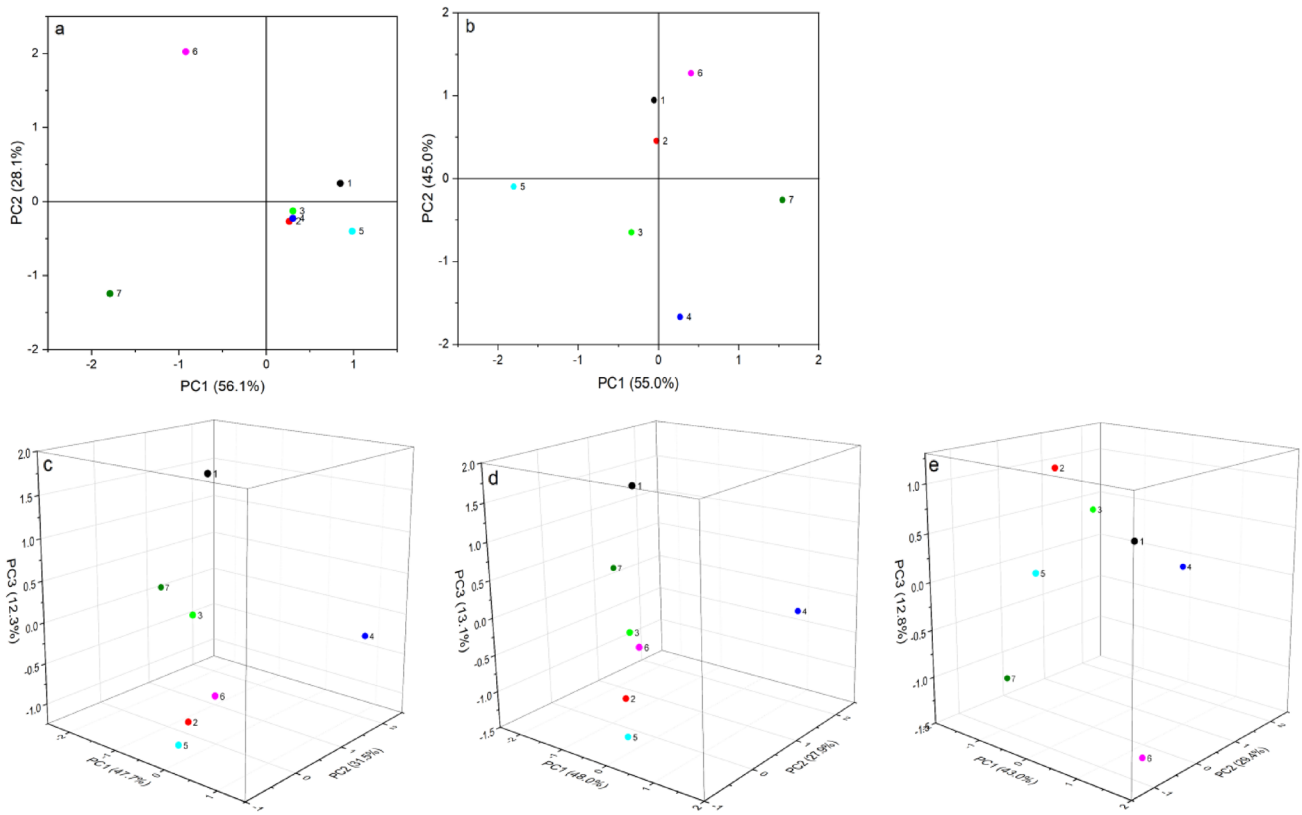

Fig. 1 Score plots of PCA. 1. *C. gigantocarpa*: 2. *C. vietnamensis*: 3. *C. osmantha*: 4. *C. nanyongensis*: 5. *C. oleifera*: 6. *C. semiserrata*: 7. *C. amplexicaulis*. **a** Fatty acid data; **b** Squalene and tocopherol data; **c** Fatty acid and tocopherol data; **d** Fatty acid and squalene data, **e** Volatile substance data

C. oleifera, *C. semiserrata*, and *C. amplexicaulis* were significantly different from *C. gigantocarpa*, *C. vietnamensis*, *C. osmantha*, and *C. nanyongensis*, which showed a close association among themselves. There was no significant difference in fatty acids among different varieties in the same region, consistent with the results of LiuH et al. [19]. The results of camellia oil classification by fatty acid content using HCA analysis are shown in Fig. 2a. Clearly, *C. amplexicaulis* differed from the other samples. However, *C. osmantha* and *C. nanyongensis* were classified into one category, consistent with the PCA result. In HCA, *C. vietnamensis* and *C. semiserrata* were classified into one category; thus, using only the fatty acid content to identify different varieties of camellia oil may be insufficient.

Characterization and discrimination of camellia oils based on squalene and tocopherol data

This study also considered squalene and tocopherols, which are the main unsaponifiable components in camellia oils (Table 2). In particular, squalene was detected in all camellia oil samples. As shown in the table, the squalene content in several camellia oil varieties is quite different. For instance, *C. semiserrata* has the highest squalene content (421.36 mg/kg), whereas *C. oleifera* has the lowest (161.60 mg/kg). This result is consistent with the results of Zhong et al. [18] but different from those of Zeng et al. [20], possibly because of the variety and origin of camellia seeds. Among the tocopherol isomers, α-tocopherol is the most important and the only tocopherol type detected in our samples. Except for *C. nanyongensis* and *C. amplexicaulis*, all oil samples have α-tocopherol. The α-tocopherol content in *C. oleifera* oil (25.78 mg/100 g) was considerably higher than in other camellia seed oils [9].

PCA results (Fig. 1b) showed that the identification ability of squalene and tocopherol was better than that of the other components. No overlap was noted among the seven varieties of camellia seed oil. HCA divided the camellia oils into two categories: *C. osmantha*, *C. nanyongensis*, and *C. oleifera* in one category; *C. gigantocarpa*, *C. semiserrata*, *C. amplexicaulis*, and *C. vietnamensis* in the other. Among them, *C. semiserrata* and *C. amplexicaulis* were in one category, *C. gigantocarpa* and *C. vietnamensis* in another, *C. osmantha* and *C. nanyongensis* in one category, and *C. oleifera* in the other; this categorization is also relatively weak in distinguishing different varieties.

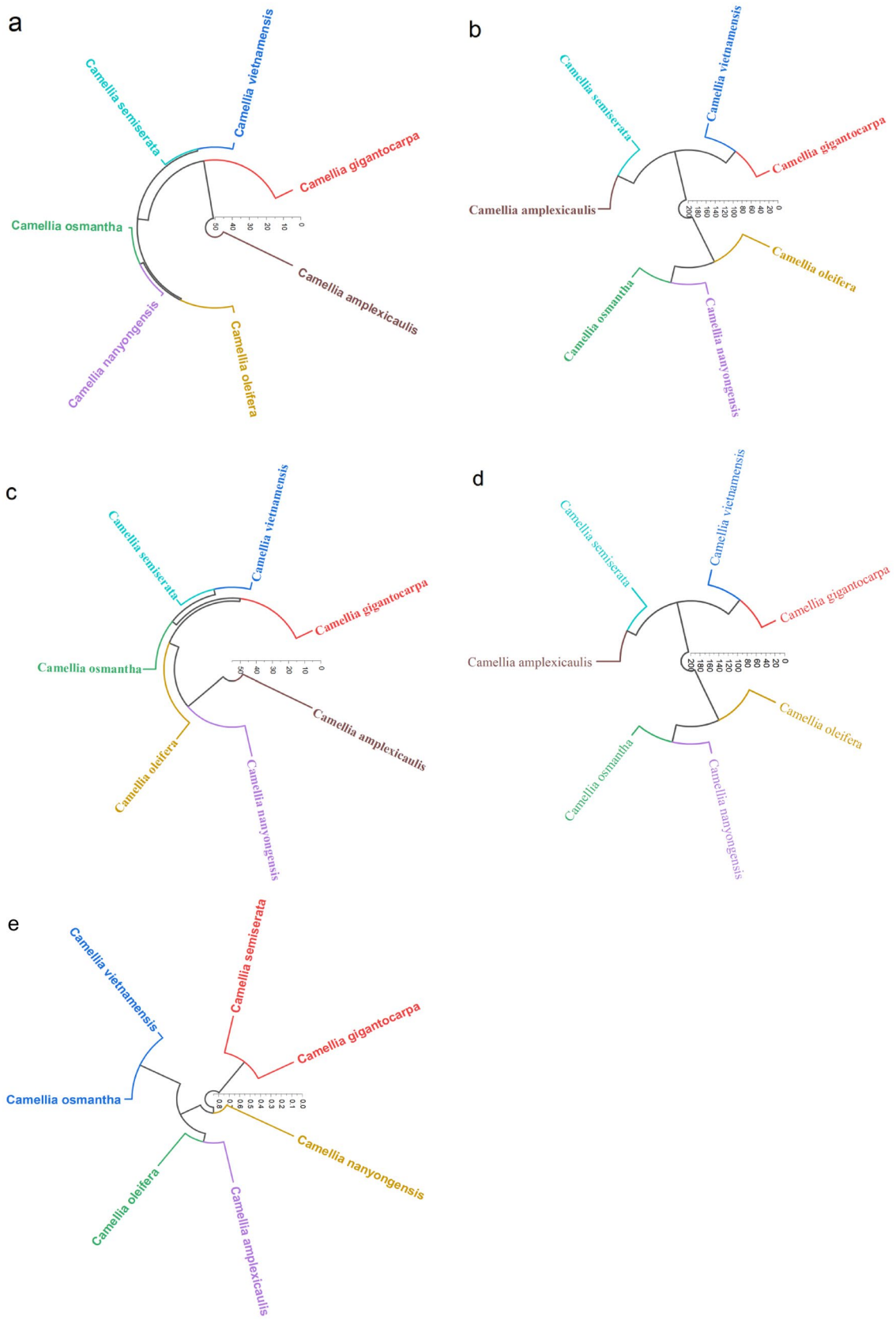

Fig. 2 HCA dendrogram. **a** Fatty acid data; **b** Squalene and tocopherol data; **c** Fatty acid and tocopherol data; **d** Fatty acid and squalene data, **e** Volatile substance data

Table 2 Squalene and tocopherols content in camellia oils

Camellia oils	Squalene content(mg/kg)	Tocopherol content (mg/100 g)				
		α-Tocopherol	β-Tocopherol	γ-Tocopherol	δ-Tocopherol	Total
C. gigantocarpa	364.41 ± 0.05	19.84 ± 0.06	ND	ND	ND	19.84 ± 0.06
C. vietnamensis	333.32 ± 0.08	16.33 ± 0.05	ND	ND	ND	16.33 ± 0.05
C. osmantha	234.59 ± 0.12	11.28 ± 0.11	ND	ND	ND	11.28 ± 0.11
C. nanyongensis	210.84 ± 0.14	ND	ND	ND	ND	ND
C. oleifera	161.60 ± 0.03	25.78 ± 0.17	ND	ND	ND	25.78 ± 0.17
C. semiserata	421.36 ± 0.11	18.60 ± 0.09	ND	ND	ND	18.60 ± 0.09
C. amplexicaulis	403.07 ± 0.04	ND	ND	ND	ND	ND

The notation "ND" was used to indicate that the substance was not detected at the limits of quantitation. To facilitate chemometric analysis, the content% of samples with the result "ND" was considered 0

[a]Data are expressed as mean ± standard deviation (SD)

Characterization and discrimination of vegetable oils based on the combination of fatty acids and squalene/tocopherol data

We also examined the ability of the combination of fatty acids and squalene/tocopherol in distinguishing camellia oil varieties. As expected, its identification ability is considerably higher than that of fatty acids alone. Compared with Fig. 1a, c and d reveal significant differences between *C. vietnamensis*, *C. osmantha*, and *C. nanyongensis*; the fatty acid and tocopherol combination of *C. vietnamensis* and *C. oleifera* were similar, while the fatty acid and squalene combination of *C. osmantha* and *C. semiserrata* were similar; the identification effect of such combinations was significantly lower than that of squalene and tocopherol combinations. The classification results of different camellia oil varieties by fatty acid and tocopherol combination are clearly better than the other data. In Fig. 2c, all camellia oils are divided into two categories, and *C. amplexicaulis* is evidently different from the other samples; each critical value can distinguish the camellia oil of one variety. The HCA dendrogram of fatty acid and squalene combination was similar to that of squalene and tocopherol combination.

Characterization and discrimination of camellia oils based on volatile substance data

Volatile substances are also an important sensory characteristic of camellia seed oil; however, few scholars have compared different varieties. However, although the differences in data between different varieties of camellia oil in the same region may not be obvious in terms of the contents of fatty acids, squalene, tocopherol, etc., the contents of trace volatile substances can be used to distinguish samples, which we aimed to explore in this study. Indeed, previous data have shown that it is necessary to combine the data of fatty acids, squalene, and tocopherol to clearly distinguish between several samples. This study identified 155 volatile substances, including aldehydes, carboxylic acids, alkanes, ketones, alcohols, esters, alkenes, ethers, and other compounds, and their contents are shown in Table 3. In *C. amplexicaulis*, 26 volatile substances were detected, including seven other compounds, except ethers and esters. This variety had the least number and types of volatile components, but its aldehyde content (73.52%) was the highest in all varieties. *C. osmantha* and *C. oleifera* have the same number and type of volatile substances (49 substances and 8 types), but they differ in acid content, which was higher in *C. oleifera* (12.34%), and alkane content, which was higher in *C. osmantha* (2.91%). In *C. nanyongensis*, we detected 56 volatile substances of 9 types, with only ethers being detected. The alcohols in *C. nanyongensis* were the highest in all samples. Furthermore, *C. gigantocarpa* has the most number of volatile substances ($n = 68$), among which ketones are the highest. Compared with *C. gigantocarpa*, *C. semiserrata* has fewer volatile substances ($n = 63$) but has the highest content of alkenes (24.76%). The next highest number of volatile substances was found in *C. vietnamensis* ($n = 58$), which also has the most number of heterocyclic compounds (27.94%). According to previous studies [21–24], aldehydes and heterocyclic compounds mainly originate from the oxidation and Maillard reaction of unsaturated fatty acids, and alcohol compounds are the intermediate products of the Strecker degradation pathways. Among all samples, except *C. semiserrata*, aldehydes and heterocyclic compounds accounted for more than 50% of the total volatile substances; alcohols also have a significant proportion (except *C. amplexicaulis*). Combining Tables 1 and 2, *C. amplexicaulis* has a very low content of unsaturated fatty acids, and *C. semiserrata* has the highest alkene content, indicating that the oxidation degree is low and may be protected by squalene and tocopherols.

Table 3 Volatile substance content in camellia oils

Types	Name	Retention time(min)	Volatile substances content(%)						
			C. gigantocarpa	*C. vietnamensis*	*C. osmantha*	*C. nanyongensis*	*C. oleifera*	*C. semiserata*	*C. amplexicaulis*
Aldehydes	Propanal, 2-methyl-	1.648	ND	0.23	ND	ND	ND	0.25	ND
	Butanal, 3-methyl-	2.268	4.45	1.7	2.45	0.15	4.68	1.54	ND
	Butanal, 2-methyl-	2.365	0.65	0.6	1.08	ND	1.3	0.55	ND
	Pentanal	2.728	ND	0.56	0.63	0.45	0.41	ND	3.59
	Hexanal	4.859	3.98	ND	ND	9.38	9.66	4.92	17.69
	2-Hexenal, (E)-	6.809	ND	ND	ND	ND	ND	ND	0.28
	2-Hexenal	6.828	ND	ND	0.26	ND	ND	ND	ND
	Heptanal	9.267	ND	6.21	6.19	ND	3.75	1.7	8.46
	2-Heptenal, (E)-	12.974	0.66	0.73	0.85	0.79	0.81	0.45	5.98
	Benzaldehyde	13.349	0.9	0.77	0.56	ND	1.99	0.85	ND
	Octanal	16.953	0.59	13.02	16.24	25.18	3.86	1.3	10.3
	Benzeneacetaldehyde	20.714	8.47	3.95	4.66	ND	3.28	4.14	ND
	2-Octenal, (E)-	22.396	ND	ND	1.32	ND	0.54	ND	1.41
	Nonanal	27.67	ND	11.46	13.14	8.99	9.02	6.14	22.27
	2-Nonenal, (E)-	34.255	ND	1.3	0.36	0.36	ND	ND	ND
	Decanal	40.123	ND	0.21	0.43	0.53	ND	ND	ND
	2,4-Nonadienal, (E,E)-	41.197	ND	ND	ND	0.2	ND	ND	ND
	2-Decenal, (E)-	44.928	ND	0.55	0.81	2.14	0.29	ND	3.07
	UNDecanal	27.613	6.69	ND	ND	ND	ND	0.13	ND
	2-UNDecenal	48.232	ND	0.17	0.24	1.55	ND	ND	0.47
Total			26.39	41.46	49.22	49.72	39.59	21.97	73.52
Carboxulic acids	Acetic acid	1.765	5.78	3.32	2.79	0.2	7.56	5.62	1.71
	Propanoic acid, 2-methyl-	3.575	0.45	0.12	ND	ND	0.44	0.52	ND
	Butanoic acid	4.305	ND	ND	ND	ND	ND	ND	1.13
	Butanoic acid, 3-methyl-	6.279	2.31	0.53	0.54	ND	1.24	0.69	ND
	Butanoic acid, 2-methyl-	6.748	ND	ND	ND	ND	1.01	0.77	ND
	Pentanoic acid	8.444	ND	ND	ND	ND	ND	ND	3.04
	Hexanoic acid	15.741	0.12	0.71	ND	ND	1	0.51	4.95
	Heptanoic acid	24.885	0.08	ND	ND	ND	ND	ND	ND
	Octanoic acid	36.846	ND	0.56	ND	1.06	0.37	ND	ND
	Nonanoic acid	45.501	ND	0.4	0.27	0.7	0.72	0.41	ND
Total			8.74	5.64	3.6	1.96	12.34	8.52	10.83
Others	Azetidine, 1-nitroso-	1.847	6.13	5.11	ND	ND	11.8	4.24	ND
	Trichloromethane	2.011	ND	0.15	ND	0.09	ND	0.07	ND

Table 3 (continued)

Types	Name	Retention time(min)	Volatile substances content(%)						
			C. gigantocarpa	*C. vietnamensis*	*C. osmantha*	*C. nanyongensis*	*C. oleifera*	*C. semiserata*	*C. amplexicaulis*
	Furan, 2,3-dihydro-5-methyl-	2.434	ND	ND	ND	0.07	ND	ND	ND
	Silanediol, dimethyl-	2.516	ND	0.52	ND	ND	ND	ND	ND
	Ethyl(dimethyl)ethoxysilane	3.092	ND	ND	ND	0.1	ND	ND	ND
	Butanenitrile, 3-methyl-	3.221	0.43	0.49	0.39	ND	0.77	0.23	ND
	4-Amino-3-hydroxytetrahydrothiophene 1,1-dioxide	3.976	2.26	ND	ND	ND	ND	ND	ND
	cis-2,3-Epoxyoctane	4.86	ND	11.45	14.56	ND	ND	ND	ND
	2H-Pyran-3(4H)-one, dihydro-	5.031	0.5	0.38	0.12	ND	0.45	ND	ND
	Pyrazine, methyl-	5.673	1	0.57	0.25	ND	1.27	1.02	ND
	Silane, ethoxytriethyl-	5.819	0.48	ND	ND	ND	ND	ND	ND
	Furfural	5.969	1.93	1.15	0.23	ND	2.04	0.58	ND
	1.2-diacetylhydrazine	6.546	0.16	0.48	0.45	ND	0.63	0.32	ND
	2-Furanmethanol	6.843	3.04	1.24	ND	ND	2.29	0.91	ND
	Pyrazine, 2,5-dimethyl-	9.847	1.72	0.75	0.74	ND	2.03	2.9	ND
	Formamide, N,N-diethyl-	11.19	ND	ND	ND	ND	ND	ND	0.57
	Dihydro-3-methylene-5-methyl-2-furanone	12.125	0.24	ND	ND	ND	ND	ND	ND
	2(3H)-Furanone, dihydro-5-methyl-	12.389	0.27	ND	ND	ND	ND	ND	ND
	Silane, triethylmethoxy-	13.665	0.44	ND	ND	0.38	ND	0.46	0.58
	O-Butylisourea	14.195	ND	ND	ND	ND	ND	ND	4.1
	1-[(Trimethylsilyl)oxy]propan-2-ol	14.82	2.75	2.44	ND	ND	ND	ND	ND
	Silane, [3-(2,3-epoxypropoxy) propyl]ethoxydimethyl-	14.865	ND	ND	2.33	5.24	ND	ND	3.18
	Furan, 2-pentyl-	15.818	ND	1.79	1.74	ND	0.47	ND	ND
	7-Oxabicyclo[2.2.1]Heptane, 1-methyl-4-(1-methylethyl)-	17.887	0.68	0.47	ND	ND	ND	1.58	ND
	2,5-FuraNDione, 3,4-dimethyl-	19.404	ND	ND	ND	ND	ND	ND	1.16
	4-Methyl-5H-furan-2-one	20.14	0.71	ND	ND	ND	ND	ND	ND
	Ethanone, 1-(1H-pyrrol-2-yl)-	22.765	0.14	ND	ND	ND	ND	ND	ND
	Cyclotrisiloxane, hexamethyl-	25.09	ND	ND	ND	1.43	ND	ND	ND
	Furan, 2-butyltetrahydro-	26.501	ND	0.29	0.53	1.95	ND	ND	ND
	2,5-DihydroxyBenzaldehyde, 2TMS derivative	28.758	0.98	ND	ND	ND	ND	0.43	ND

Table 3 (continued)

Types	Name	Retention time(min)	Volatile substances content(%)						
			C. gigantocarpa	*C. vietnamensis*	*C. osmantha*	*C. nanyongensis*	*C. oleifera*	*C. semiserata*	*C. amplexicaulis*
	2,5-Pyrrolidinedione, 1-ethyl-	31.111	ND	ND	ND	0.25	ND	ND	ND
	4H-Pyran-4-one, 2,3-dihydro-3,5-dihydroxy-6-methyl-	31.575	0.19	ND	ND	ND	ND	ND	ND
	Disiloxane, 1,1,3,3-tetramethyl-1,3-bis[3-(oxiranylmethoxy)propyl]-	32.808	0.17	ND	ND	ND	ND	ND	ND
	Butyl tiglate 2-methyl-1-	38.699	ND	ND	ND	ND	ND	0.49	ND
	benzimidazolemethanol, 5(or 6)-chloro-alpha-methyl-	44.378	ND	ND	ND	ND	0.26	ND	ND
	2(3H)-Furanone, 5-butyldihydro-	44.395	ND	0.35	ND	0.39	ND	ND	ND
	PhosphonoAcetic acid, 3TMS derivative	44.906	ND	ND	ND	ND	0.48	ND	ND
	Anethole	45.894	ND	ND	ND	ND	0.18	ND	ND
	(3S,3aS,8aR)-6,8a-Dimethyl-3-(prop-1-en-2-yl)-1,2,3,3a,4,5,8,8a-octahydroazulene	49.552	ND	ND	ND	ND	ND	0.2	ND
	(1R,4R,4aS,8aR)-4,7-Dimethyl-1-(prop-1-en-2-yl)-1,2,3,4,4a,5,6,8a-octahydronaphthalene	50.143	ND	ND	ND	0.69	ND	ND	ND
	1,3-Cyclohexadiene, 5-(1,5-dimethyl-4-hexenyl)-2-methyl-, [S-(R*,S*)]-	50.75	ND	ND	ND	0.32	ND	ND	ND
	Naphthalene, 1,2,3,5,6,8a-hexahydro-4,7-dimethyl-1-(1-methylethyl)-, (1S-cis)-	51.161	0.47	0.31	0.49	0.14	0.74	0.35	ND
Total			24.69	27.94	21.83	11.05	23.41	13.78	9.59
Alkanes	*n*-Hexane	1.854	ND	ND	2.42	0.16	ND	ND	0.37
	Hexane, 2,2-dimethyl-	2.647	ND	ND	ND	ND	ND	ND	0.85
	Heptane	2.784	ND	ND	0.49	0.57	0.32	0.06	ND
	Bicyclo[3.1.1]Heptane, 6,6-dimethyl-2-methylene-, (1S)-	14.556	ND	ND	ND	ND	ND	0.28	ND
	Dodecane	39.636	0.31	0.16	ND	ND	ND	ND	ND
	Tetradecane	49.026	ND	0.23	ND	ND	ND	ND	ND

Table 3 (continued)

Types	Name	Retention time(min)	Volatile substances content(%)						
			C. gigantocarpa	*C. vietnamensis*	*C. osmantha*	*C. nanyongensis*	*C. oleifera*	*C. semiserata*	*C. amplexicaulis*
	Benzene, 1-(1,5-dimethyl-4-hexenyl)-4-methyl-	50.517	ND	ND	ND	1.22	ND	ND	ND
	Benzene, 1,1'-(1,1,2,2-tetramethyl-1,2-ethanediyl)bis-	50.889	ND	ND	ND	0.16	ND	ND	ND
Total			0.31	0.39	2.91	2.11	0.32	0.34	1.22
Ketones	2-Propanone, 1-hydroxy-	2.323	ND	0.65	0.42	ND	ND	0.89	ND
	2,3-Pentanedione	2.697	0.37	ND	ND	ND	ND	0.21	ND
	Acetoin	2.867	1.13	0.38	0.27	0.19	0.77	0.82	ND
	Dimethadione	7.050	0.36	ND	ND	ND	ND	ND	ND
	2-Propanone, 1-(acetyloxy)-	7.355	0.26	ND	ND	ND	ND	ND	ND
	4-Cyclopentene-1,3-dione	8.077	0.14	ND	ND	ND	ND	ND	ND
	2-Heptanone	8.554	ND	0.24	0.43	0.3	0.42	0.15	0.56
	5-Hydroxy-4-octanone	8.657	1.1	ND	ND	ND	ND	ND	ND
	1,2-Cyclopentanedione	10.49	0.15	ND	ND	ND	ND	ND	ND
	2,5-Hexanedione	10.692	0.19	ND	ND	ND	ND	ND	ND
	2-Nonanone	26.032	ND	0.15	ND	ND	ND	ND	ND
	aR-Turmerone	53.157	ND	ND	ND	1.75	ND	ND	ND
	Tumerone	53.223	ND	ND	ND	0.38	ND	ND	ND
Total			3.7	1.42	1.12	2.62	1.19	2.07	0.56
Alcohols	cis-3-Methylcyclohexanol	2.800	ND	ND	ND	0.24	ND	ND	ND
	1-Butanol, 3-methyl-	3.32	3.37	0.35	0.2	ND	1.58	0.57	ND
	Propylene Glycol	3.365	ND	ND	ND	0.64	ND	ND	ND
	1-Butanol, 2-methyl-	3.385	ND	ND	0.36	ND	2.18	3.42	ND
	2-Propanol, 1-isopropoxy-2-methyl-	3.846	1.91	ND	ND	ND	ND	0.4	ND
	1,2-Butanediol	3.859	ND	0.15	ND	ND	ND	ND	ND
	1-Butanol, 2-methyl-, acetate	3.973	ND	3.16	ND	ND	3.33	ND	ND
	1-Propanol, 2-methyl-	3.982	ND	ND	ND	ND	ND	2.96	ND
	1-Pentanol	3.989	ND	ND	ND	4.54	ND	ND	0.75
	CH3C(O)CH2CH2OH	4.426	ND	ND	ND	1.34	ND	ND	ND
	1,3-Butanediol, (S)-	4.499	9.37	ND	ND	ND	ND	9.19	ND
	2,3-Butanediol	4.576	ND	1.5	0.77	5.26	4.81	ND	ND
	Propane-1,3-diol, 2-methyl-	5.029	ND	ND	ND	ND	ND	0.6	ND
	1-Hexanol	7.566	0.75	2.31	2.27	ND	0.95	1.32	ND
	1-Butanol, 3-methyl-, acetate	7.904	0.12	ND	ND	ND	ND	0.45	ND
	2-Butanol, 1-methoxy-	8.302	0.4	ND	ND	ND	ND	ND	ND

Table 3 (continued)

Types	Name	Retention time(min)	Volatile substances content(%)						
			C. gigantocarpa	*C. vietnamensis*	*C. osmantha*	*C. nanyongensis*	*C. oleifera*	*C. semiserata*	*C. amplexicaulis*
	1,2-Pentanediol	11.699	1.29	ND	ND	1.49	ND	ND	ND
	2-Heptenol	13.971	ND	ND	ND	1.66	ND	ND	ND
	1-Heptanol	14.181	0.33	4.88	5.25	3.32	1.61	0.64	ND
	3-Nonyn-2-ol	15.73	0.62	ND	ND	ND	ND	ND	ND
	2-Decen-1-ol, (E)-)	19.494	ND	ND	ND	0.73	ND	ND	ND
	Benzyl alcohol	19.887	0.11	ND	ND	ND	ND	ND	ND
	2-Nonen-1-ol, (E)-	23.582	0.28	ND	ND	1.04	ND	ND	ND
	Phenylethyl Alcohol	28.196	0.91	ND	ND	ND	0.56	1.01	ND
	2-Nonyn-1-ol	34.184	0.1	ND	ND	ND	ND	ND	ND
	(1R,2R,4S,6S,7S,8S)-8-Isopropyl-1-methyl-3-methylenetricyclo[4.4.0.02,7]decan-4-ol	49.908	ND	ND	ND	0.46	ND	ND	ND
Total			19.56	12.35	8.85	20.72	15.02	20.56	0.75
Esters	MethoxyAcetic acid, butyl ester	3.376	ND	0.19	ND	ND	ND	ND	ND
	Acetic acid, pentyl ester	3.993	ND	ND	3.38	ND	ND	ND	ND
	Acetic acid, ethoxyhydroxy-, ethyl ester	4.308	4.59	ND	ND	ND	ND	ND	ND
	(Z)-Ethyl 2-methylbut-2-enoate	8.965	ND	ND	0.52	ND	ND	3.2	ND
	Heptanoic acid, 2-methyl-2-butyl ester	9.432	ND	ND	ND	0.16	ND	ND	ND
	Butyrolactone	9.696	1.1	0.18	ND	ND	0.8	1.53	ND
	4-Methylcyclohexanol acetate	15.794	ND	ND	ND	2.67	ND	ND	ND
	2-Butenoic acid, 2-methyl-, 2-methylpropyl ester, (Z)-	21.643	ND	ND	ND	ND	ND	0.3	ND
	Hexanoic acid, 2-propenyl ester	22.431	ND	ND	ND	1.77	ND	ND	ND
	Butyric acid, 4-pentadecyl ester	46.377	ND	ND	0.12	ND	ND	ND	ND
	trans-.alpha.-Bergamotene	49.278	ND	ND	ND	ND	ND	1.89	ND
Total			5.69	0.37	4.02	4.6	0.8	6.92	0
Alkenes	1-Octene	4.641	ND	ND	ND	ND	ND	ND	0.55
	Tetrachloroethylene	5.139	ND	ND	ND	0.25	ND	ND	0.36
	Styrene	8.697	ND	ND	ND	ND	ND	0.69	ND

Table 3 (continued)

Types	Name	Retention time(min)	Volatile substances content(%)						
			C. gigantocarpa	*C. vietnamensis*	*C. osmantha*	*C. nanyongensis*	*C. oleifera*	*C. semiserata*	*C. amplexicaulis*
	Bicyclo[4.2.0]octa-1,3,5-triene	9.234	ND	ND	0.33	ND	1.9	ND	ND
	.alpha.-Pinene	11.331	0.58	0.38	0.38	ND	0.42	0.79	ND
	Cyclohexene,3-(1-methylpropyl)-	15.773	ND	ND	ND	ND	ND	0.8	ND
	3-Carene	17.305	0.12	ND	ND	ND	ND	0.36	ND
	1,3-Cyclohexadiene, 1-methyl-4-(1-methylethyl)-	18.101	ND	0.18	ND	ND	ND	ND	ND
	Cyclohexene, 1-methyl-4-(1-methylethylidene)-	18.111	ND	ND	ND	ND	ND	0.9	ND
	p-Cymene	18.83	1.6	1.37	0.86	ND	0.73	3.37	ND
	D-Limonene	19.305	4.79	3.5	2.65	ND	2.21	13.69	ND
	.gamma.-Terpinene	22.239	0.45	ND	ND	ND	ND	0.9	ND
	Cyclohexene, 3-methyl-6-(1-methylethylidene)-	25.232	1.48	1.18	0.68	ND	ND	2.72	ND
	Benzene, 1-methyl-3-(1-methylethenyl)-	25.76	0.09	ND	ND	ND	ND	ND	ND
	UNDecane	27.409	0.22	ND	ND	1.56	ND	ND	1.41
	5-Tridecene, (E)-	36.057	ND	ND	ND	0.58	ND	ND	ND
	1-UNDecene, 8-methyl-	47.559	ND	ND	ND	0.46	ND	ND	ND
	Caryophyllene	49.422	ND	ND	ND	0.4	ND	ND	ND
	Epizonarene	50.808	ND	ND	ND	0.22	ND	ND	ND
	.alpha.-Muurolene	50.843	0.31	0.19	0.23	ND	0.38	0.24	ND
	trans-Calamenene	51.206	0.32	0.2	ND	ND	0.38	0.24	ND
	cis-Calamenene	51.218	ND	ND	0.23	ND	ND	ND	ND
	Cyclohexene, 3-(1,5-dimethyl-4-hexenyl)-6-methylene-, [S-(R*,S*)]-	51.22	ND	ND	ND	0.78	ND	ND	ND
	Cyclohexene, 4-[(1E)-1,5-dimethyl-1,4-hexadien-1-yl]-1-methyl-	51.444	ND	ND	ND	ND	ND	0.06	ND
Total			9.96	7	5.36	4.25	6.02	24.76	2.32
Ethers	Ethyl 1-cyclohexen-1-yl ether	44.216	ND	ND	ND	0.93	ND	ND	ND
Total			0	0	0	0.93	0	0	0

The notation "ND" was used to indicate that the substance was not detected at the limits of quantitation. To facilitate chemometric analysis, the content% of samples with the result "ND" was considered 0

[a]Data are expressed as mean ± standard deviation (SD)

PCA results (Fig. 2) showed that the identification ability of volatile substances was better. No overlap was observed among the seven varieties of camellia seed oil.

Conclusion

In conclusion, this study detected fatty acids, squalene, tocopherols, volatile substances, and other characteristic compounds in camellia seed oil by HPLC, GC, and GC–MS. Seven varieties of camellia seed oil were identified and classified by PCA and HCA. Fatty acids, squalene, tocopherol, and volatile substances could be used as means to distinguish camellia seed oil varieties. However, different characteristic compounds showed different discrimination for camellia seed oil varieties, and even a single component could not clearly distinguish these camellia seed oil varieties. For example, the ability of fatty acids to identify *C. vietnamensis*, *C. osmantha*, and *C. nanyongensis* was significantly lower than fatty acid and tocopherol combination or fatty acid and squalene combination. Notably, the volatile substances are also excellent in identifying different varieties of camellia oils; thus, they can be used as evidence for the identification of different camellia oil varieties and as the basis for the fine processing of camellia oil.

Acknowledgements This research work was financially supported by the Science and Technology Major Project of Guangxi (No. AA20302021-6), and the project was funded by Guangxi Key Laboratory of Special Non-wood Forests Cultivation & Utilisation (No. JA-22-04-02).

Author contribution All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Guiqing LI, Yan Jiang and Xingyue Tang. The first draft of the manuscript was written by Xingyue Tang, Jihua Guan and Mi Qiu and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability The data will be made available upon reasonable request to the corresponding author.

Declarations

Conflict of interest The authors declare no conflicts of interest.

Ethical approval This research work was financially supported by the Science and Technology Major Project of Guangxi (No. AA20302021-6), and the Project funded by Guangxi Key Laboratory of Special Non-wood Forests Cultivation & Utilisation (No. JA-22-04-02). This article does not contain any studies with human or animal subjects.

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